Object-Oriented Programming

Collections and Sorting

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SDU

Part 0:
The `Collections` Namespace

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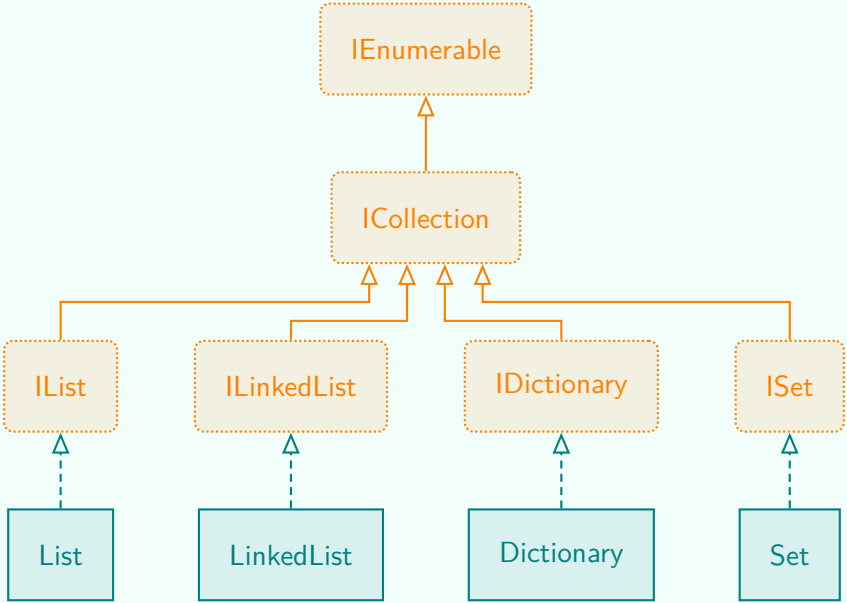
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The namespace also contains a number of classes that implement these interfaces (with the described behaviour).

The `Collections` Namespace ▷ Hierarchy



Part 1:
The `ICollection` Interface

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`ICollection` interface contains a number of method prototypes. These include prototypes for:

1. `Add` is used to ensure that the collection contains a specific element.
2. `Remove` is used to remove *a single* reference to a specific object.
3. `Clear` is used to remove all elements from the collection.

The `ICollection` Interface ▷ Types of Collections

There are a number of different collections in the `Collections` namespace:

- `ISet` Data structures containing a group of unique elements.
- `IList` Data structures containing an ordered group of elements.
- `ILinkedList` Data structures containing a chain of elements.
- `Stack` Data structure that orders elements according to last-in/first-out.
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By implementing the `ICollection` interface!

Part 2:
Lists

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- It is a *dynamic* array. Dynamic because it automatically adjusts its size so that it appears that there is no wasted space.
- Belongs to the .NET Framework (which we will discuss later).
- The user has control over where in the sequence an element is inserted.

Lists ▷ The `IList<T>` Interface

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This means that elements are arranged in a sequence, and that we can determine what should be located where:

- Properties:
  - `int Count`
  - `T[] Item`
- Methods:
  - `void Insert (int index, T element)`
  - `void RemoveAt (int index)`
  - `int IndexOf (T element)`

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There are several classes in the `Collections` namespace that implement this interface, but the most important one is `List<T>`.

Lists ▷ The `List<T>` Class

Stores elements in an array. Ensures that the array is replaced with a new (larger) array if the capacity of the original is exceeded, or a new (smaller) one if there ends up being too much wasted space.

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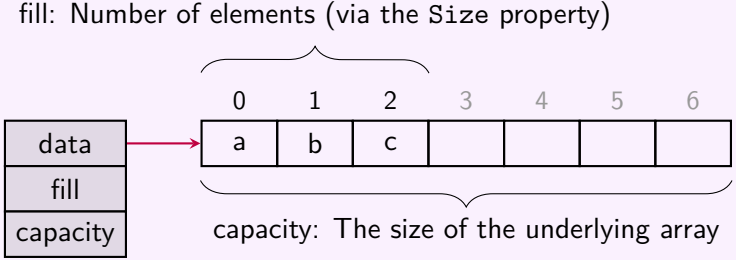
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List<Student> students = new List<Student>();
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List<Student> students = new List<Student>();
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Only objects of the specified type (and subtypes thereof) can be stored in a `List`.

If you try to add an object of a different type (one that is not more specific), the program cannot be compiled.

Lists ▷ Usage ▷ `Object`-Parameterization

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However, this is rarely done. Why is that?

Because then we only have access to the methods defined on `Object` on the elements of `list`.

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Example:

```
List<string> list = new List<string>();
```

Lists ▷ Usage ▷ Important Instance Methods

```
void Add (T obj);            //  add item to the end of the list

bool Contains (T obj);       // returns true if obj is in the list

void Insert (int N, T obj); // insert element at N'th position, and shift

void RemoveAt (int N);       // remove element at N'th position
                             // (subsequent elements are moved to
                             // fill the gap)

bool Remove (T obj);         // if a reference to obj is found in the list
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These methods use `Equals`

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With a `for` loop (and access through the index):

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List<int> list = GetRandomList():

for (int i=0 ; i<list.Count ; i++) {
  list.Items[i] += i;
}
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Or with a foreach loop (without access to the index):

```
foreach (int item in list) {
  Console.WriteLine(item);
}
```

Part 3:
Linked Lists

Linked Lists ▷ Class `LinkedList`

Stores elements in a *chain*, where elements *easily* can be added and removed from the beginning or end.

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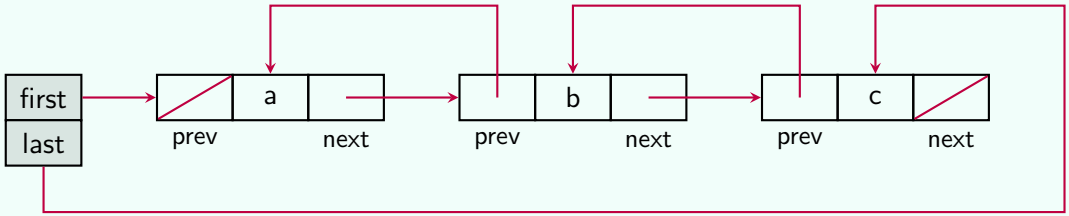
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Part 4:
Comparison

Comparison ▷ Introduction

A number of the things that we might be interested in doing with `Collection` objects require us to have a flexible way of comparing two elements.

Example: Sorting a list involves making a lot of comparisons.

Comparison ▷ The `IComparable` Interface

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public int CompareTo (T other);
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Comparable defines one method:

```
public int CompareTo (T other);
```

What does it do?

- Return value is negative when the current object is less than `other`.
- Return value is positive when the current object is greater than `other`.
- Return value is 0 when the current object is at level with `other`.

Comparison ▷ The IComparable Interface ▷ Example (1/2)

```
public class Person : IComparable<Person> {
  private int age;

  public Person (int age) {
    this.age = age;
  }

  public int GetAge () {
    return this.age;
  }

  public int CompareTo (Person? t) {
    return this.age - (t==null ? 0 : t.GetAge());
  }

  public override String ToString () {
    return "" + this.age;
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```

Alternative way to write CompareTo:

```
public int CompareTo (Person? t) {
  int otherAge = (t==null
                  ? 0
                  : t.GetAge());

  if (this.age > otherAge){
    return 1;
  } else if (this.age < otherAge){
    return -1;
  }
  return 0;
}
```

Comparison ▷ The IComparable Interface ▷ Example (2/2)

```
public class Program
{
  public static void PrintList (List<Person> persons) {
    Console.Write("List<Person> [");
    for (int i=0 ; i<persons.Count ; i++) {
      Console.Write((i==0 ? "" : ",")+persons[i]);
    }
    Console.WriteLine("]");
  }

  public static void Main ()
  {
    List<Person> persons = new List<Person>();
    persons.Add(new Person(72));
    persons.Add(new Person(4));
    persons.Add(new Person(17));
    PrintList(persons);
    persons.Sort();
    PrintList(persons);
  }
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  }
}
```

Output:

```
List<Person> [72,4,17]
List<Person> [4,17,72]
```

If classes had to implement the `IComparable` interface in order to be sorted, this would impose a few limitations:

- Since we have not created all the classes ourselves, it is not always possible for us to ensure that they implement `IComparable`, and thus it would not be possible to sort elements of arbitrary types.
- We would only be able to sort in one way.

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For instance, using the `IComparer`-interface, we can define a class whose purpose is to compare `string`-objects.

The `IComparer` interface helps us overcome the challenges posed by the `IComparable`:

- We are not dependent on being able to change the definition of the classes whose objects we want to compare.
- We can implement as many `IComparer`-classes as we want for a given type.

Comparison ▷ IComparer ▷ Example

```
public class DoubleComparer : IComparer<double>
{
  public int Compare (double s1, double s2)
  {
    return (int)(s1 - s2);
  }
}

public class Program
{
  public static void Main ()
  {
    List<double> stringList = new List<double>();
    // fill it up
    stringList.Sort(new DoubleComparer());
  }
}
```

Part 5:
Maps

Maps ▷ Introduction

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- This code is generated via a call to `GetHashCode` which is defined on `Object`.
- **Note:** If you override `Equals`, then you must also (consider) overriding `GetHashCode`: Two objects that are "equal" must have the same hash code.

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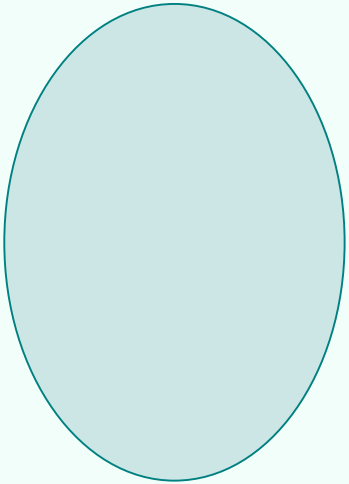
Rules:

- One key can have one associated value.
- Multiple keys can be associated with the same value.
- You can ask whether a key is associated with a value.

Maps ▷ Mapping from Keys to Values

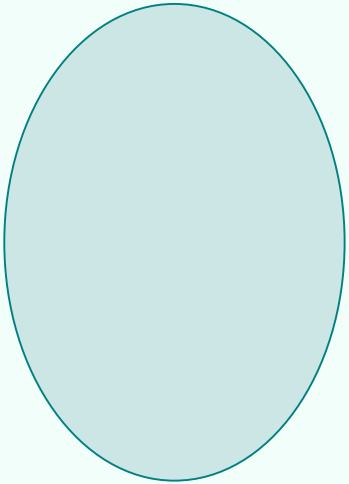
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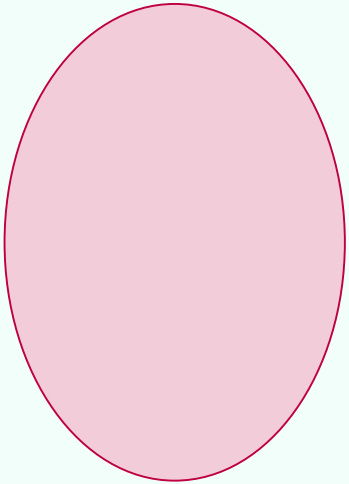


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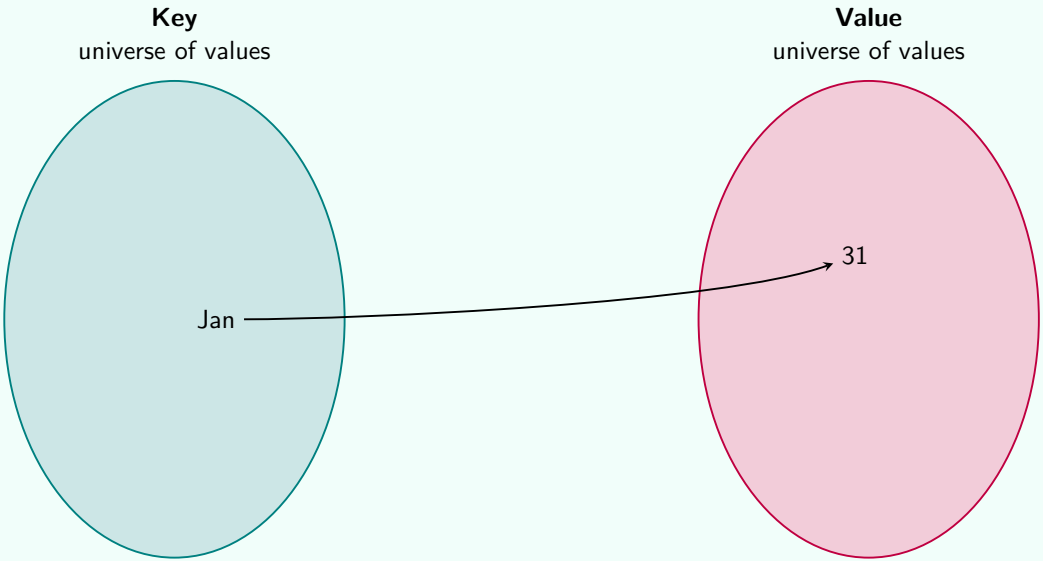
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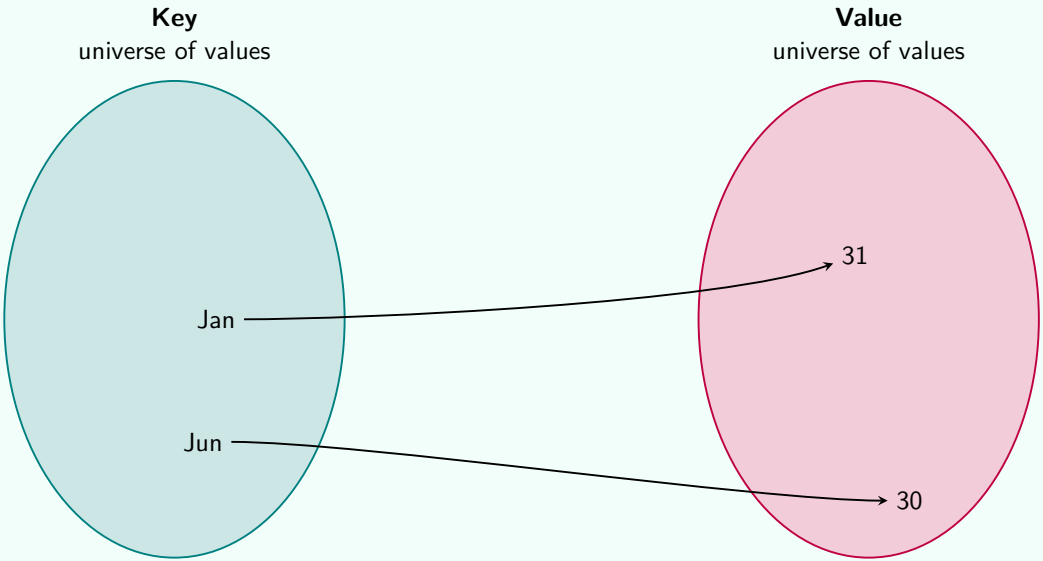
Value
universe of values



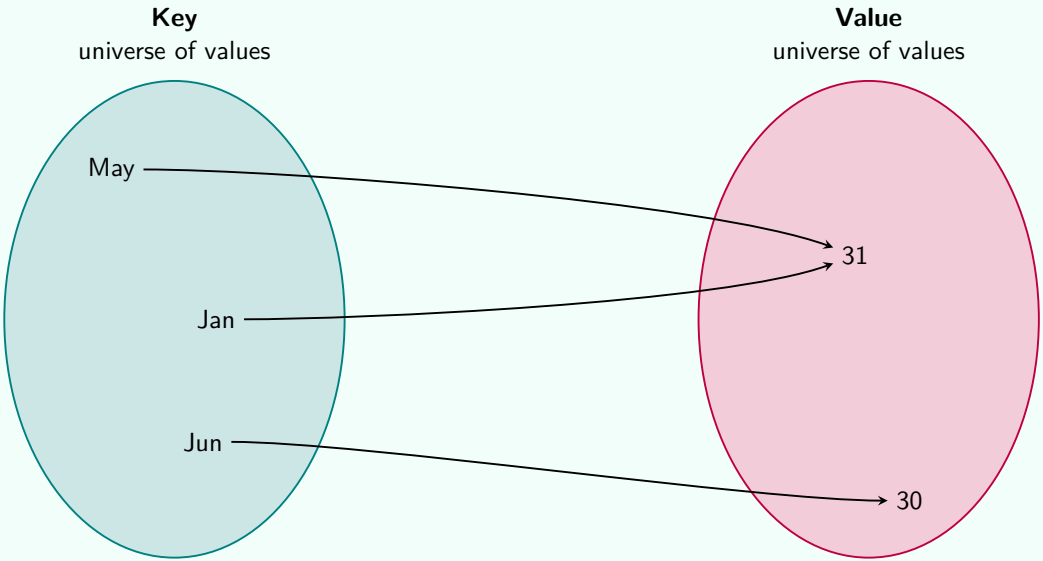
Maps ▷ Mapping from Keys to Values



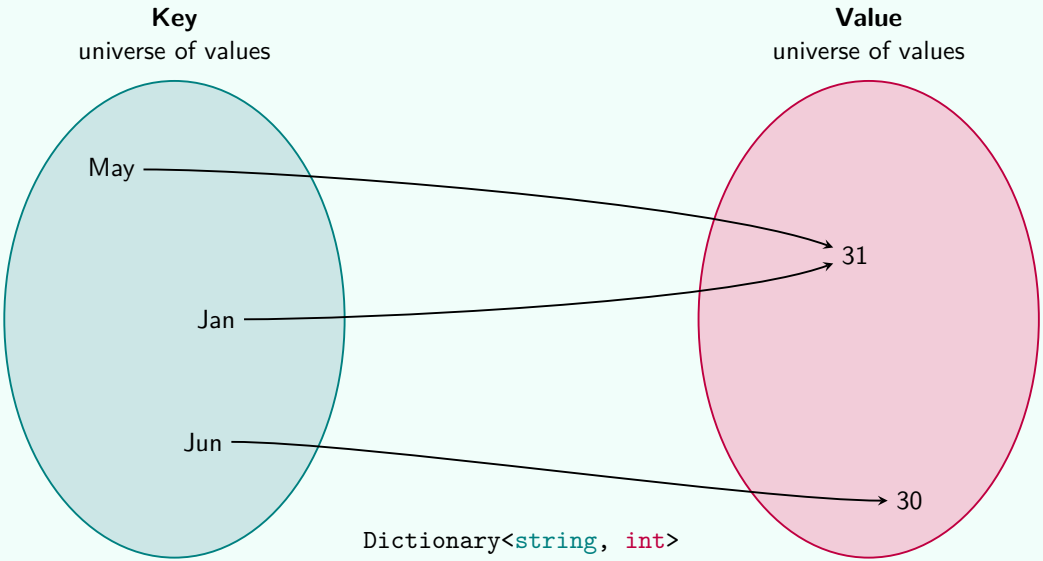
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Maps ▷ The IDictionary Interface

`IDictionary` defines a number of methods, including:

- `void Add (K key, V value)`
- `void Clear ()`
- `bool ContainsKey (K key)`
- `bool Remove (K key)`

`IDictionary` has a number of properties, including:

- `int Count`
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Since a key can only point to a single value, `Add` will update existing *bindings*.

Maps ▷ The `Dictionary` Class

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```
Dictionary<string, Person> map = new Dictionary<string, Person>();
Person p = new Person();
map.Add("Aslak", p);
Person fromMap = map["Aslak"];
```

Part 6:
Sets

Sets ▷ Introduction

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The `HashSet` class implements this interface and – like `Dictionary` – uses a hash code to distinguish between object identities.

> *Therefore, the previous rule about overriding `GetHashCode` also applies if it should be possible to place an object of the type in question in a `HashMap`. And since it is very easy to overlook, and we do not know how our classes will be used in the future, we should always adjust `GetHashCode` when we tinker with `Equals`.*

Sets ▷ The `ISet` Interface

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`ISet` defines, among other things, the following instance methods (which are *destructive*):

- `bool Add (T element)`
- `void Clear ()`
- `bool Contains (T element)`
- `bool IsSubsetOf (IEnumerable other)`
- `bool IsSupersetOf (IEnumerable other)`
- `bool Overlaps (IEnumerable other)`
- `void IntersectWith (IEnumerable other)`
- `void UnionWith (IEnumerable other)`

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Part 7:
Sorting

Sorting

Sorting

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But we need to understand what a sorting algorithm might look like and why it works. We will use "insertion sort" as an example of a sorting algorithm.

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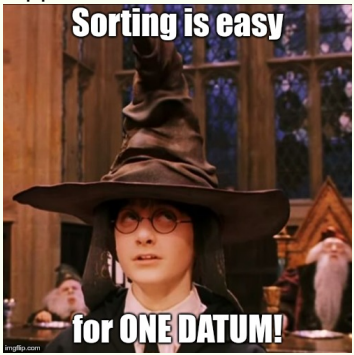
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Sorting ▷ Insertion Sort ▷ Introduction

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Q: Why is the first element skipped?
A: Because a sequence consisting of a single element is, by definition, sorted!

In iteration i, it is ensured that all elements up to the current one are sorted:

1. Remember the value of the i'th element.
2. Move all elements between the correct position for the i'th element and position i one position forward.
3. Save the original value of the i'th element in the correct position.

Sorting ▷ Insertion Sort ▷ Algorithm

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   Break out of the loop by jumping to point 3.
3. `list[j+1]` ← `element`

https://www.youtube.com/watch?v=ROalU379l3U

Sorting ▷ Insertion Sort ▷ Implementation

```
// sort the array arr into increasing order
public int[] InsertionSort (int[] arr) {
  // iterate over all items that may need relocation
  for (int itemsSorted=1 ; itemsSorted<arr.Length ; itemsSorted++) {
    int temp = arr[itemsSorted]; // the item to be inserted
    int loc = itemsSorted - 1;   // start at end of the ordered list

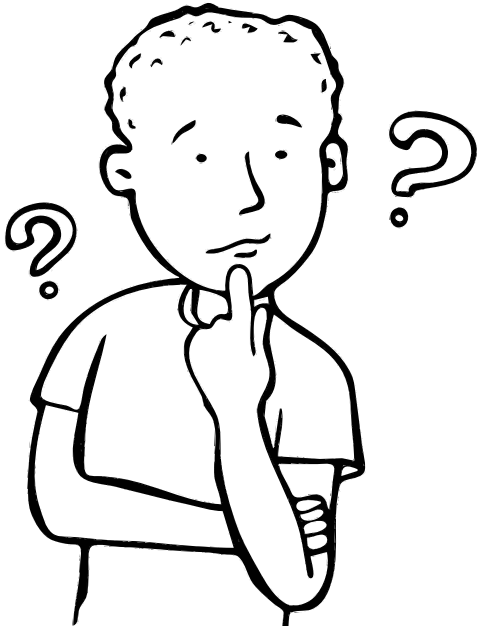
    // move all items larger than temp one spot to the right
    while (loc>=0 && arr[loc]>temp) {
      arr[loc+1] = arr[loc]; // bump item from arr[loc] up to loc+1
      loc = loc - 1; // go on to next location
    }

    // put temp in last vacated space
    arr[loc+1] = temp;
  }

  return arr;
}
```


YOU'RE SLOW
QUICK SORT
MERGE SORT
HEAP SORT
SELECTION SORT
MERGE SORT
QUICK SORT
HEAP SORT
SELECTION SORT
US TOO
SELECTION SORT
BUBBLE SORT
INSERTION SORT

Questions?



https://openclipart.org/detail/238087/boy-thinking-of-question